MAKERERE UNIVERSITY

College of Computing and Information Sciences

School of Computing and Informatics Technology

Department of Networks

BSE4104 - Emerging Trends in Software Engineering

AI Native and Agentic Engineering Capstone Project

GROUP J EVENING

WEEK 2: FOUNDATION MODEL

SME PROCUREMENT PREPARATION AGENT

NAMES	REGISTRATION NUMBERS
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Work Completed Against Weekly Objectives

- Selected and documented the foundation model with a formal Model Selection Note covering capability, cost, latency, privacy and access considerations.
- Integrated the model into a working baseline application where we had python script using the OpenAI API.
- Created prompt specification v1.0 for the low stock detection and reorder recommendation capability, including role, task, context, constraints, output format and failure behaviour.
- Developed a ready to use prompt text and successfully demonstrated baseline model interaction with synthetic inventory data.
- Designed and documented 10 test cases covering normal, boundary, edge, missing data and safety scenarios.
- Prepared the evaluation table ready for execution and recording of results.
- Versioned the prompt (v1.0 baseline established; ready for v1.1 improvements after test execution).
- Updated the GitHub repository with all new artefacts under /prompts, /docs, and /docs/evaluation.
- Maintained the ClickUp board with Week 2 tasks assigned, tracked, and closed.

The team now has a tested, versioned baseline model interaction that correctly identifies low-stock items and recommends reorder quantities while respecting the defined safety boundaries.

Key Engineering Decisions and Rationale

DECISION	RATIONALE
Selected GPT-4o-mini	Best balance of cost, speed, structured output quality and ease of integration for a student project. Higher capability models were judged unnecessary for the current analytical task.
Strict, constrained prompt design	Prevents hallucination of stock levels, prices or unauthorised actions. Forces the model to stay within the defined role and output format.
Deterministic data + AI explanation	Inventory numbers remain under code control for example the model only reasons and formats. This maintains clear AI vs deterministic boundaries.

Temperature set to 0.2	Reduces randomness and improves consistency of structured table output across test runs.
Early creation of 10 diverse test cases	Allows systematic evaluation before moving to RAG or tools following the course principle of establishing a solid baseline first.

Challenges Encountered and Current Response

Challenge	Current Status
Ensuring the model consistently follows the exact table output format	Tightened the prompt instructions and added explicit format examples. Will re-test after any v1.1 changes.
Handling missing fields gracefully	Added explicit failure behaviour rules in the Prompt Specification. Test cases TC06 - TC08 specifically target this.
Balancing prompt length vs clarity	Kept the system prompt focused and moved data into the user message.
Slight delay in completing full test execution	Evaluation table is ready; actual runs and Pass/Fail recording will be finalised at the start of Week 3.

Repository and Task Management Evidence

GitHub Repository: https://github.com/Nicholas-Muwonge/SME-Procurement-Agent

ClickUp Board:
https://app.clickup.com/1200420000004171/v/f/1200420000011824/1200420000005776

Plan for Week 3

Focus: Context Engineering and RAG

- Assemble a controlled corpus (inventory policies, sample supplier terms or reorder guidelines – 10–50 documents/records).
- Record provenance of all sources.
- Implement ingestion, chunking, indexing and retrieval.
- Construct model context from retrieved evidence and display sources.
- Create at least 15 RAG test questions (answerable, partially answerable, unanswerable).
- Document at least three retrieval/grounding failures and their causes.

- Deliver: Corpus/Source Register, RAG architecture diagram, working RAG pipeline, 15-case evaluation results and Week 3 Progress Report.

Prepared by: Nicholas Muwonge on behalf of the team